Virtual University of Pakistan

(Zoo503)

(Paleontology and Zoogeography)

Final term MCQs

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Topic 81-160

A network of collagen fibers on which needlelike crystals of hydroxyapatite (a form of apatite, CaPO4) accumulate The Skeleton

Bone

Cartilage are the type of tissues which are progressively mineralized by:

Apatite

The first vertebrates had a _____ skeleton

Cartilaginous

The _____ appears in the early embryo as a strip of cells where the backbone will develop Jawless Fishes

neural crest

in _____, commonest vertebrates were the conodont animals

Late Cambrian and Ordovician

_____ Were jawless, generally armored

Ostracoderms

_____ are the extinct genus of jawless fishes that lived in Devonian period.

Hemicyclaspis

They occur as phosphatic toothlike microfossils, termed elements.

Conodonts

Conodonts are divided into ____ groups

3

_____more complex, with cones, bars and blades

Euconodonts

The three main morphotypes of conodonts are:

Coniform, Ramiform, Pectiniform

• The basal vertebrates, lacked jaws, jaws probably evolved during the_____

Ordovician

• Oldest jawbearing fishes were the_____

Placoderms

chondrichthyans were the first _____ like cartilaginous fishes

Shark

Small fishes, mostly in the range _____ in length.

50–200 mm

The _____ are Ancestors of most fishes today

Actinopterygians

The first good evidence of tetrapods were _____ long

0.6 and 1 m

_____ are 4000 species of mainly small animals that live in or close to water

Half-way land animals

The oldest fossil form, with reduced limbs, is _____ in age

Jurassic

A part of amniotic egg that collects waste.

Allantois

_____ are Distinguished by the pattern of openings in the side of the skull

Late Carboniferous

The three major skull patterns seen in amniotes:

anapsid, diapsid and synapsid

All modern amniotes produce the ____ egg

Cleidoic

The oldest Turtles and relatives were _____ eaters

small insect

First synapsids were Grouped loosely as "_____"

Pelycosaurs

Reptiles typically have ____ bones in the lower jaw

Six

In mammals, there is a single bone in the lower jaw known as:

the dentary

As compared to the three tiny bones in human ear, reptiles have only one, known as:

stirrup or stapes

Dinosaurs famously ruled the Earth for _____ of the Mesozoic

160 Myr

The mammals achieved great diversity and abundance only after the extinction of:

Dinosaurs

A group, includes the dinosaurs, pterosaurs, crocodilians and birds.

Archosaurs

_____ has a lightweight body, Narrow hatchet-shaped skull, a long narrow wing

Proficient flapping flyers

First true lizards are:

Lepidosaurs

The theropods and sauropodomorphs share the primitive _____ pattern

reptilian hip

All other dinosaurs share a unique hip pattern in which the pubis has swung back and runs parallel to:

Ischium

fish-shaped animals, adapted to life in the sea

The ichthyosaurs

Marine reptile _____ uses snake like neck for hunting the fish in water.

Plesiosaurs

The plesiosaur, _____ Swam by beating their paddles in a kind of "flying" motion.

Cryptoclidus

The most famous fossils of Oldest known bird were Found in Upper Jurassic sediments in southern Germany in

1861

Examples of Flightless ratites and ancestors of water birds

penguins and birds of prey

Small insect eaters in the Late Triassic and Early Jurassic Remained small through most of the:

Mesozoic

A classic example of an adaptive radiation due to development of teeth

Marsupials & Placentals

_____ are early primates

Humans

• Homo sapiens means:

wise person

primates have Good _____ vision

binocular

_____ primates radiated in the Paleocene, Eocene and Oligocene

Basal

_____ in Marmosets and spider monkeys are used as extra limbs in swinging through the trees.

Prehensile Tails

• The _____ of Southeast Asia are the most primitive living apes

Gibbons

_____ of the DNA of Gorillas, chimpanzees and humans is identical.

>94%

In_____, two French teams announced human fossils that were 6 Ma.

2001 and 2002

The famous skeleton of a female P. afarensis from Ethiopia, called ____ by its discoverer

Lucy

Lucy's brain is small, only ____ for a height of 1–1.2 m

415 cm^3

Another human species, H . erectus had Brain size of _____ in a body up to 1.6 m tall

830– 1100 cm^3

Homo erectus

Truly modern humans, H . sapiens, may have arisen as much as ____ years ago

400,000

_____ had large brains (on average, 1400 cm3)

Neandertals

_____, which concentrates on macroscopic (visible with the naked eye) plant remains

Paleobotany

Palynology is the study of_____.

fossil pollen and spores

Trace fossils are named on the basis of_____

shape and ornamentation

The study of trace fossils is often called _____

Ichnology

structures formed in hard substrates, such as limestone, shells or wood, for the purpose of protection

Borings

non-fecal pellets are regurgitation pellets of_____

birds and reptiles

Two models for the diversification of marine invertebrate life over the past 600 myr of good quality fossil records.

The Empirical Model, The Bias Simulation Model

the plateau appears clearly only in the _____ data compilation.

family-level

A single equilibrium level is easier to understand in terms of a:

global equilibrium model

The Modern _____ radiated dramatically over the last 100 myr

"fauna"

_____ means change with improvement.

Progress

_____ are observations of the appearance and disappearance of species.

Patterns

_____ are hypotheses that seek to explain the patterns.

Processes

The replacement of brachiopods by bivalves is a famous example of:

Biotic replacements

First living organisms' bacteria:

(3500 Ma)
Oldest multicellular eukaryotes: reported from rocks _____ in Canada
1200 Ma
A process of reproduction in which a new individual arises by the fusion of two sex cells, or gametes, produced by different individuals
Sexual reproduction
Composed of many kinds of cells – muscle cells, nerve cells, epithelial cells and so on
Differentiated cells
the study of the distribution of species and ecosystems in geographic space and through geological time.
Biogeography
_____ are ideal locations because they allow scientists to look at habitats that new invasive species have only recently colonized
Islands
He noted many of the mammals of North America, such as bears, deer, squirrels, hedgehogs and moles, were found also in Eurasia • Travelled via Alaska, when climates were warmer
Georges Buffon
the founder of plant geography
German Alexander von Humboldt:
Augustin de Candolle of Geneva and Lamarck published a map showing France divided into _____ regions with different ecological conditions.
five floristic
who was the first to classify the world's flora and show the results on maps?
Danish botanist Joakim Schouw
(113) Evolution – A Dangerous Idea!
French Revolution of ____ led to a flowering of French science
1789
Power of the Church was broken:
18th century
_____ of Natural History, became a powerhouse of ideas and debate in Europe.
New National Museum
In 1802, _____ suggested that • 'Lower' organisms might also be found earlier in time and that they might gradually change into the 'higher' forms
Jean- Baptiste Lamarck
Geoffroy suggested evolutionary homologies and links between such widely different animals as:
fish and cephalopods
Darwin had spent the _____ after his voyage in detailed research on many other areas of biology that provided evidence for evolution
40 years

_____ is the central paradigm of the earth sciences and Theory of evolution is of the biological sciences

Plate tectonics

____ was the first to make a world map

Engler

The system of plant regions accepted today is very similar to that of:

Engler

“all species had been created within the area in which they are found today”. • This was first formalized by:

British ornithologist Philip Sclater

Presented his theory of continental drift in (1912)

Pangaea Alfred Wegener

Croizat published his ideas in the 1950s and 1960s, his major presentation being his 1958 book:

Panbiogeography

Plants in High levels of heat and moisture

Megatherms

Plants in Moderate levels of heat and moisture

Mesotherms

plants in Low levels of heat and moisture

Microtherms

James Dana divided the _____ into several different zones based on mean minimum temperature.

surface waters

_____ in 1856 published Five depth zones and 25 faunal provinces along the coasts of the continents

Edward Forbes

three British zoologists Assembled _____ of the distributions of fishes based on the patterns of distribution of 27 families.

30 maps

Sheer variety and volume of the works published by:

Alfred Wallace Candolle

New Guinea contributes only _____ of the world’s land area and some 10% of its species of terrestrial organism

3%

A species that exists as a series of subspecific forms is termed A _____ Species

Polytypic

All chimpanzee subspecies lie to the north or east of this river

River Congo

A _____ may be deterred from crossing an area of grassland; or a marsh organism may fail to travel across dry habitats to reach the next area of wetland.

forest species

_____ that leaves its forest- floor log, is exposed beetles, shrews and insectivorous birds.

Insect

_____ Dive for their food and bring vegetable matter to the surface from greater depths.

Coots

The coots are messy eaters, and it is not difficult for gadwall to move in and collect some of the loot. This behavior is called

Klepto-parasitism

Patterns of Distribution

• Species may evolve in one region, spread to other locations and then become extinct in all but a restricted area where it survives. Species restricted in this way are said to be

Endemic

the two major factors influencing the degree of endemism in an area:

Isolation, Stability

Australia is indeed rich in endemics, many of which have a long geological history. This kind of 'fossil endemism' is called

Palaeo-endemism

Anything that tends to make it more difficult for a species to live, grow or reproduce in its environment may prove to be a:

limiting factor

Any regular change in physical or chemical conditions through space that creates a sequence of replacement of one species by another, among both animals and plants.

zonation

the predators can always turn to alternative food species if the numbers of their usual prey should be reduced by climatic or other influences. This is termed.

Prey Switching

_____ reduce the rate of growth of populations because of their negative influence on general fitness, survival and fecundity.

Parasites

The removal of certain species could create effects far in excess of what may originally have been expected. Influential species of this kind are known as.

Keystone Species

The loss of a _____ can cause an avalanche of local extinctions.

Keystone Species

Microscopic plankton are capable of changing the depth at which they live, depending on conditions, and this can be regarded as a form of:

vertical migration

_____ of a species can be measured by its geographical distribution

Success

The term _____ is used for animals that play a particular role in the community, such as plant-sucking insects or insectivorous birds

guild
To cope with the physical demands of their environment
Species adapt
species tolerating one another, or becoming dependent on one another
Co- evolution
Pointing out that no two species have quite the same needs.
Individualistic concept
_____ of certain species from an ecosystem would have little or no effect on the functioning of that ecosystem
Removal
A possible alternative model is based on the supposition that all species are equally important to:
ecosystem function
certain species play key roles in the ecosystem, and when they are lost there is a sudden drop in the capacity of the ecosystem to function. This is sometimes referred to as:
The Rivet Hypothesis
_____ is Defined in terms of inertia, or resistance to change.
Stability
Woody savannas – trees > 2 m high, cover only _____ of the land surface
30–60%
Variations in climate also result from the pattern of movement of
air masses
The 'trade winds', found in both the Northern and Southern Hemispheres, meet in the region of the equator, and this is known as the:
intertropical convergence zone (ITCZ)
Approximately _____ species of organism have been described
1.8 million
Of the bacteria, only about ____ species have so far been described
4000
_____ have always been most abundant in the low-latitude (tropical) regions.
Angiosperms
Greater extinction rates in the higher latitudes is the instability of climatic conditions over the past ____ years
2 million
High-latitude organisms have _____ than those from the low latitudes.
broader geographical ranges
General feature of biogeography first pointed out by ____ in the 1970s
E.H. Rapoport
Katherine Smith and James Brown examined the diversity of ____ as one proceeds deeper
fish

The Amazon basin contains approximately _____ flowering plant species, whereas equivalent areas in Africa and in South-East Asia contain only about 40,000 each.

90 000

All the land surface area of the hotspots together comprise only about ____ of the Earth's terrestrial total

1.4%

The idea that continents could fragment and move across the face of the planet was first suggested by the German meteorologist _____ in 1912

Alfred Wegener

The surface of the Earth, known as the lithosphere, is occupied by a number of areas known as:

tectonic plates

The _____ move as well as the continents, the study of their movements is known as plate tectonics rather than continental drift

ocean floors

The fossil record suggests that all the early amphibian and reptile groups evolved in the continent called

Euramerica

_____ are all the result of the break-up and dispersal of Gondwana

Southern continents

mainly the result of the appearance and disappearance of land or sea barriers within North America and Eurasia

Northern continents

Apart from humans and the aerial bats, only the ____ spread naturally as far as Australia

rats

Human beings probably arrived _____ years ago and brought the domestic dog (the ancestor of the dingo) about 3500 years ago

60 000 to 40 000

Geological studies show that _____ is the smallest (about 1600 km2) surviving fragment of Gondwana

New Caledonia

New Zealand has Its total area, _____, makes it far larger than any other Pacific oceanic island

270 000 km2